

20260921

variational_gamma

- Reworked genome chunking to address memory issues (memory allocation during genome position sorting)
- Chunk size of 10000000 (run in parallel)
 - Chr2: 242193529 bp --> 25 chunks
 - Chr3: 198295559 bp --> 20 chunks
 - Chr5: 181538259 bp --> 19 chunks
 - Chr16: 90338345 bp --> 10 chunks
- Intermediate output files saved in `array_5388/tmp` or `meta_5668/tmp` so that even if individual jobs fail due to OOM (out-of-memory) error, progress is not lost for successfully completed segments
- Final output expected by end of day or early tomorrow

snpArcher

- snpArcher progress:
 - | | |
|-----------------------------|---|
| concat_interval_vcfs_stage | 2 |
| concat_interval_vcfs | 1 |
| variant_filtration | 1 |
| postprocess_basic_filter | 1 |
| qc_vcftools_individuals | 1 |
| postprocess_strict_filter | 1 |
| qc_subsample_snps | 1 |
| postprocess_subset_indels | 1 |
| postprocess_subset_snps | 1 |
| qc_prepare_plink_inputs | 1 |
| postprocess_drop_indel_SNPs | 1 |
| qc_plink | 1 |
| qc_setup_admixture | 1 |
| qc_admixture | 1 |
| qc_qc_dashboard | 1 |
- `concat_interval_vcfs_stage` : This job establishes ("stages") intervals for interval-specific `.vcf` files which are then concatenated, resulting in a genome-wide variant dataset.
- `concat_interval_vcfs` : Computational reassembly and concatenation of the interval-specific `.vcf` files into the combined genome-wide variant dataset

- **This is similar to CCDG's WGS-processing strategy for large population-genomic projects. Computationally intensive variant calling is performed in genomic partitions. Resulting calls are then subsequently consolidated and concatenated before applying QC and filtering.**
- `variant_filtration` : Variant-level filtration for the combined variant dataset. Note that the complete set of called variants is not necessarily the same as the final set, because variant-level quality metrics are used to identify calls that do not meet the desired quality criteria (**These criteria were chosen to match CCDG standards**).
- `postprocess_basic_filter` : Post-processing filter applied to the variant dataset. **Methodologically this is analogous to the CCDG approach as it consists of a progressive filtering strategy:**
 - variants produced by calling
 - variants passing initial QC criterion
 - variants passing stricter downstream criterion
 - final set for analysis
- `qc_vcftools_individuals` : Individual-level quality control using `vcftools` . **This is analogous to sample-level QC in CCDG.** In this way, both the sample-level and variant-level QC are similar between our `snpArcher` workflow and CCDG.
- `postprocess_strict_filter` : A stricter postprocessing filtering step to establish a more conservative analysis dataset. **This is in-line with the CCDG progressive filtering.**
- `qc_subsample_snps` : SNP subsampling as part of quality control. This is for reasons of computational feasibility.
- `postprocess_subset_indels` : Create a subset for INDEL variants
- `postprocess_subset_snps` : Create a separate subset for snps. **This matches the separation of SNPs and indels in the CCDG methodology.**
- `qc_prepare_plink_inputs` : Prepare filtered data for PLINK analysis. **This matches CCDG-style workflows, in which the initial large WGS variant dataset is converted into a standardized PLINK-compatible representation.**
- `postprocess_dropindel_SNPs` : Remove SNPs associated with INDEL-containing and overlapping variant representations. **This is analogous to CCDG methodology for constructing a snp-specific set for downstream analysis.**
- `qc_plink` : PLINK-based QC
- `qc_setup_admixture` : Quality control for analysis of SNP dataset using ADMIXTURE.

****Methodologically, this is analogous to CCDG transforming data for input to ancestry/population-structure analysis.**
- `qc_admixture` : ADMIXTURE analysis, output to `./results`
- `qc_qc_dashboard` : Quality control metrics output